

Secure examinations for 500 candidates

System design, security and delivery alternatives

Client decision presentation
Hosting and app management only
8 September 2026

The decision is where the exam must survive

Reliable internet: managed custom platform (B)

Preserves this app's workflows and central oversight. Add durable storage and test failover before launch.

Unreliable internet: centre LAN with delayed cloud sync (C)

Candidates save continuously to a local authority. The exam can continue through an internet outage.

No network during the exam: native offline client (D)

Supports the requested end-of-exam upload model, with greater endpoint risk and more reconciliation work.

Lower ownership effort: Moodle or a hosted service (F / G)

Accept a different workflow. Compare required security features and contractual capacity before choosing.

Recommendation is an engineering judgment. None of the proposed configurations has passed a 500-candidate production load test.

Scope and planning assumptions

Item	Basis used in this presentation
Concurrent load	500 active candidates, plus approximately 20 staff. 500 seeded records do not establish capacity.
Exam frequency	Assumption: 8 sessions/month, 3 hours/session. 4,000 candidate sittings/month and 48,000/year.
Application workload	100 questions/attempt. Budget 360 save events/attempt, including edits and retries.
Evidence	Optional snapshots: one 100 kB image/minute, retained 90 days. No video or automated analysis in the base custom budget.
Commercial boundary	Hosting, backups, operational management and scheduled exam support. Equipment and facility costs excluded by request.
Price geography	USD list-price reference: AWS US East and DigitalOcean standard USD plans. Cloudflare is global. India-only hosting needs a reprice and service review.

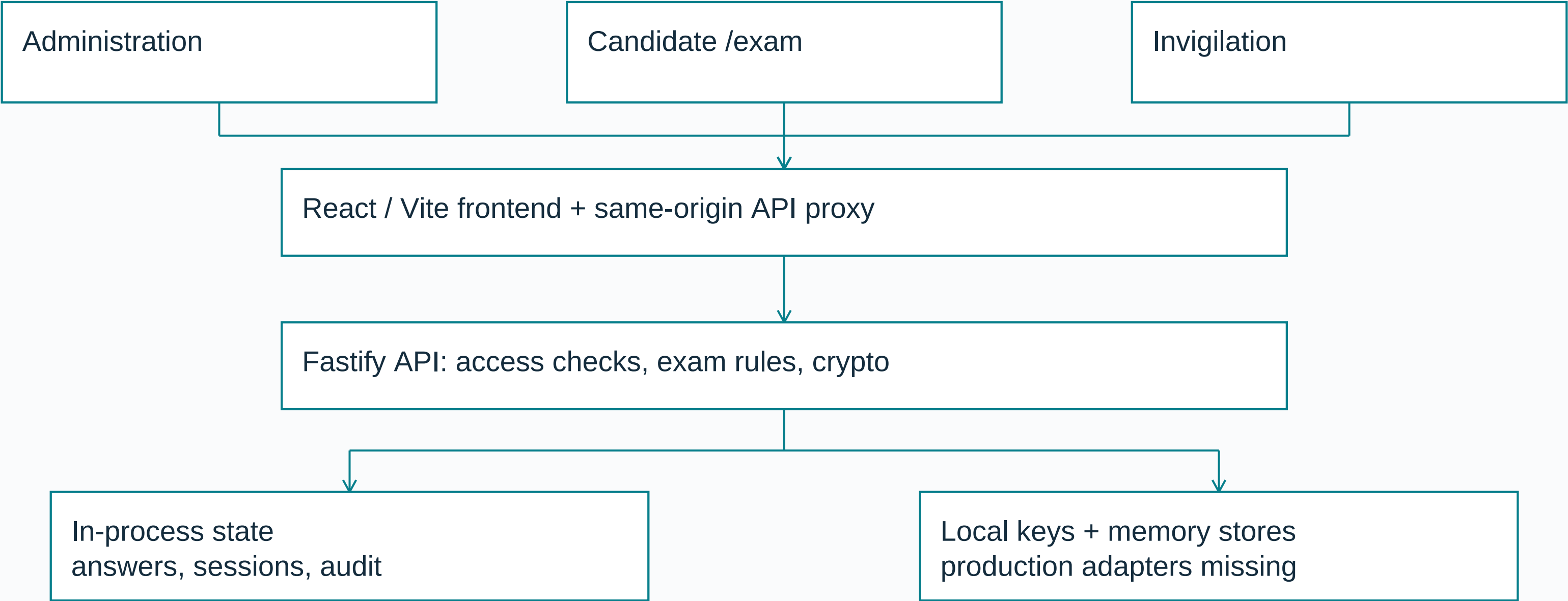
Frequency, exam duration and retention remain planning inputs, not client-approved requirements. Taxes, FX and software development are outside the recurring totals.

Assumptions embedded in the existing design

Design assumption	Evidence in the app	Alternative when it is false
Organisation controls the centre	Registered devices, network allowlist, separate exam shell	Native offline package or hosted remote assessment with a different threat model
Server remains authoritative	Server timer, question-by-question delivery, central attempt state	Local centre authority (C), or signed offline permit and local journal (D)
One backend can coordinate the demo	In-process maps, sessions, audit sequence and key provider	Shared transactional persistence before multiple API replicas
Staff resolve identity disputes	Warnings and invigilator actions with a reason	Formal exception roster and appeal procedure
Security policies can be layered	Standard, Enhanced, High and Maximum Assurance profiles	Decouple biometrics from network and key controls when privacy or access needs differ

These are reconstructed from repository code and documentation. The repository does not establish that clients approved them.

Current app: three experiences, one memory-backed API



PostgreSQL, Redis and MinIO appear in Docker Compose, but the adapter registry still instantiates memory implementations. Restart recovery is not production-ready.

Administration and publication flow

1 Author and review

Create question versions. A different person approves content. Preserve earlier versions for traceability.

2 Configure the exam

Choose timing, navigation, marking and security policy. Assign candidates and centre workstations.

3 Freeze and approve

Build the approved paper manifest. Separate assembly from publication approval. Maximum Assurance adds two approvers.

4 Seal and release

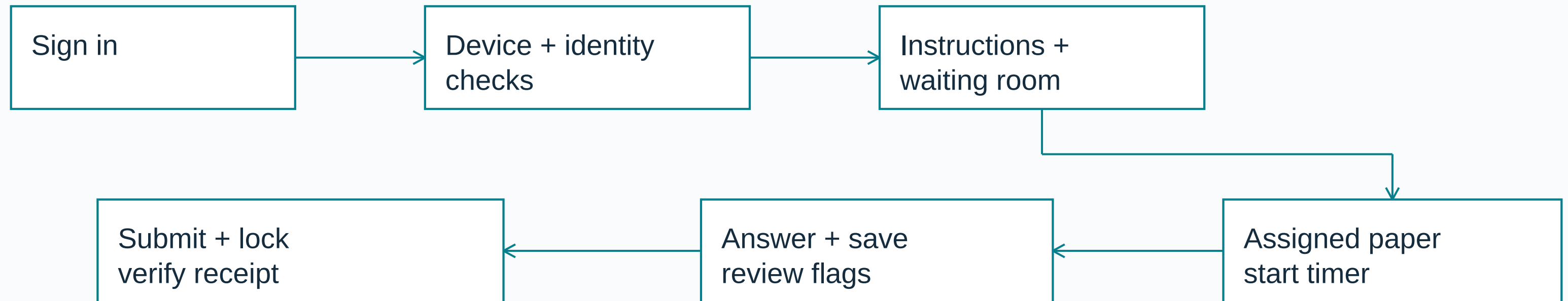
Hash, sign and encrypt. Check the release window and re-verify integrity before assigning the paper.

5 Operate and close

Review readiness, supervise attempts, verify receipts and export results with an access trail.

Production release approval must also govern key use. A UI approval flag alone cannot prevent a privileged operator from decrypting early.

Candidate flow and recovery points



Failed identity check: staff review, retry or authorised exception.

Connectivity loss: current POC retries pending changes, but cannot guarantee recovery after a browser restart.

Offline alternatives issue a local sealed receipt first, then a separate central acceptance receipt.

Question order remains attached to the same attempt after reconnection. Offline package delivery requires a new flow.

Invigilator flow keeps decisions with people

Readiness and admission

Check assigned candidates, device status and unresolved verification problems before the exam starts.

Warnings and escalation

A capture failure progresses from a warning to a prominent warning and then a policy-driven restriction.

Human review

Inspect available evidence, reverify identity or approve a documented exception. Record actor, reason and decision.

Candidate continuity

Restrictions preserve answers. Automated face or presence results must never independently terminate an attempt.

After the exam

Reconcile missing uploads, resolve incidents and provide an appeal route before final results.

POC camera capture is real where supported. Face matching, liveness and presence verdicts are simulated. Offline D cannot provide live central monitoring.

Paper protection: real cryptography, unfinished custody

Control	Current implementation	Production requirement
Content integrity	Canonical JSON + SHA-256 question and manifest hashes	Persist immutable approved versions and verify the exact released package
Authenticity	Ed25519 signatures, verified before assignment	Managed signing key, independent public-key verification and rotation
Confidentiality	AES-256-GCM with per-exam key, random nonce and manifest hash as AAD	KMS envelope keys, least-privilege decrypt and controlled release
Key separation	Separate signing and encryption material	Durable custody, recovery procedures and independent key-use logs
Candidate payload	Answer key and explanations omitted from delivery	For offline packages, omit answer keys entirely, even from locally encrypted content

The priced KMS model assumes a supported ECC signing key. Verify algorithm support and update the signature format if replacing POC Ed25519. KMS does not automatically enforce the exam schedule.

Answer security and the durable write design

Step	Protection required
Authorise	Derive candidate identity from the session. Verify attempt ownership, assigned question and allowed options.
Validate state	Check active attempt, navigation rules, permitted answer type and server deadline.
Commit atomically	One database transaction for answer, expected version, idempotency record and answer event. Unique constraints prevent duplicate attempts and replay.
Acknowledge	Return success only after durable commit. Preserve pending client data until the matching acknowledgement arrives.
Finalise	Atomically freeze final answers, compute the answer-set hash and produce a signed receipt. Reject later edits.
Recover	Replay the stored assignment. Quarantine conflicting submissions instead of silently choosing the last arrival.

Ownership and version checks exist. Current commits are map writes, not durable transactions. Production must implement atomicity, persistence and crash recovery.

Identity, session and application controls

Area	Present in code	Work before production
Authentication	script password hashing, random salt, temporary lockout	Unique credentials, staff MFA/SSO, recovery process and demo-account removal
Sessions	Signed HttpOnly cookie, SameSite=Lax, expiry checks	Secure transport, shared session store, reauthentication compatible with a full exam
Access control	Server-side roles and permissions, candidate ownership	Review centre scoping, privilege changes and least-privilege service accounts
Browser protections	CSRF token checks, exact-origin CORS, CSP, frame blocking	Review login/logout CSRF exceptions. Confirm headers on frontend responses as well as API responses
Input and logging	Zod validation, 1 MB body limit, credential redaction	Validate proxy headers, sensitive exports and log retention. Use parameterised production SQL

CSP allows inline styles, not inline scripts. CSRF uses a session-associated token. Documentation labels do not replace verification of deployed response headers.

Device and network controls need a real trust boundary

Existing demonstration	Required enforcement	Residual risk
Device registration and certificate records	Native agent, TPM-backed private key, verified device identity and revocation	A submitted workstation-code header proves no device identity
Fullscreen and readiness indicators	OS kiosk policy, app allowlisting, restricted local admin and device management	Fullscreen alone cannot block other applications
IP allowlist	Known proxy chain, firewall rules, isolated exam network	One allowed IP can represent 500 candidates and a compromised device
Synthetic attack scenarios	Edge DDoS protection, WAF rules, origin restriction and tested quotas	Public network protection cannot repair a failed centre internet link
Health checks	Verified agent state and pre-exam configuration audit	Certificates authenticate a device, not the person sitting at it

Basic cost options include an edge Pro plan, not enterprise mTLS/attestation services. Maximum Assurance integration and licences need a separate quote.

Audit, evidence and privacy controls

Area	Existing behaviour	Production completion
Audit history	Hash chain, verification, receipt anchor, no edit/delete API	Independent S3 Object Lock archive, separate credentials and external checkpoints
Evidence provenance	Server challenge, sequence and prior-evidence hash	Server validates captured evidence. Hashes cannot prove that a camera showed the true scene
Evidence access	Internal object keys and access logging	Private buckets, narrow retrieval permission, short-lived access and access reviews
Retention	Policy dates on evidence records	Scheduled deletion, retries, legal holds, backup expiry and deletion verification
Biometric privacy	Simulated identifiers and no sensitive-attribute inference	Minimise collection, publish purpose, assess provider accuracy and document an accessible exception path

An application-controlled hash chain can be rewritten with the store. Object Lock retains protected versions, so retention and deletion policies must agree.

Five production gaps directly affect a 500-seat exam

Durability and scaling

Storage adapters instantiate memory implementations. Sessions and keys disappear on restart. Multiple API instances cannot safely share this state.

Shared-IP rate limit

Default is 600 requests/minute/IP, or 10 requests/second. A centre NAT can group all 500 candidates into this budget.

Session length

Default session TTL is 60 minutes. A three-hour exam needs secure renewal or an appropriately scoped longer session.

Offline safety

Pending answers live in React refs. The candidate session store persists journey metadata, not a durable answer journal.

Trust and operations

trustProxy=true needs a restricted proxy chain. KMS, real device trust, WORM export and recovery drills remain launch work.

Findings come from source inspection, not an independent penetration test. No code changes or 500-user benchmark were performed for this presentation.

Capacity model for 500 simultaneous candidates

Traffic component	Planning calculation	Result
Answer saves	500 / 30 seconds	16.7 requests/second
Camera challenge + upload	500 x 2 / 60 seconds	16.7 requests/second
Status polling	500 / 60 seconds	8.3 requests/second
Question fetches	500 x 100 / 10,800 seconds	4.6 requests/second
Base traffic / stress target	46.3 rps plus staff and retries / roughly 3x headroom	Test 150 rps sustained and a 500-client start burst
Monthly API budget	4,000 sittings, workflows and retry margin	5 million requests; planning ceiling, not telemetry

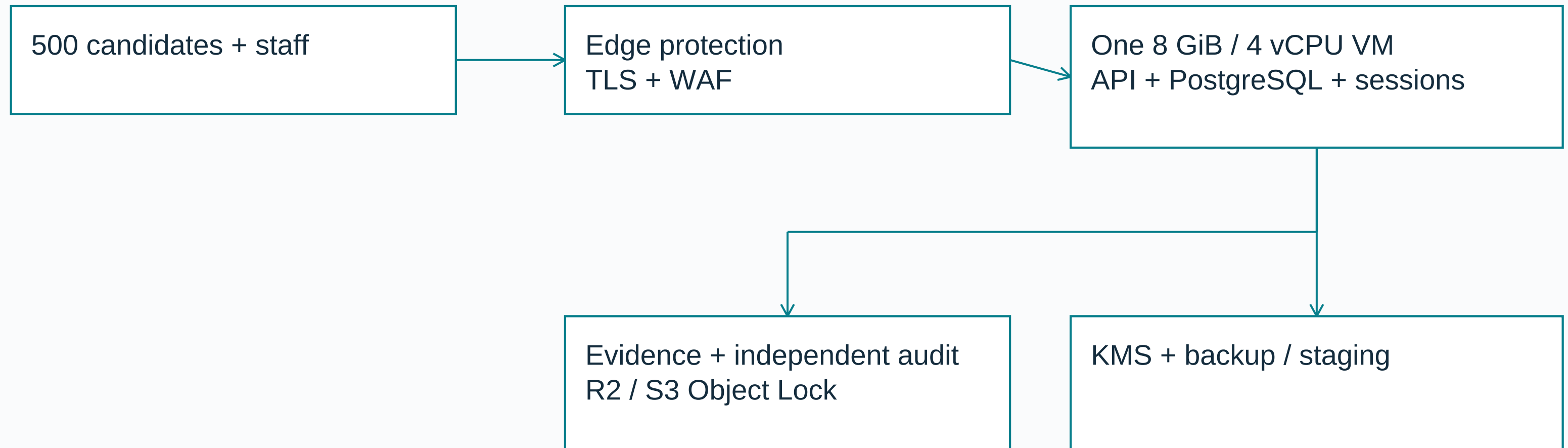
30-second saves and 60-second polling are modelling inputs. Current saves are event-driven with a 15-second pending retry sweep. Load-test cryptography and login CPU separately.

Seven alternatives for the same exam problem

Option	Design family / inspiration	Main benefit	Main tradeoff
A	Custom app on one VM	Small cloud bill	Single failure domain
B	Managed database + replicated API	Continuity and maintainability	Higher base hosting cost
C	Centre LAN + store-and-forward	Works without internet during exam	Local service operations
D	Exemplify-inspired offline client	Works without exam-time network	Endpoint trust and reconciliation
E	Cloudflare Workers + D1	Low usage-based cloud bill	Substantial backend rewrite
F	Moodle + Safe Exam Browser	Mature assessment workflows	Custom controls need plugins
G	TestInvite / managed exam SaaS	Provider manages platform	Usage charges and contract limits

This is a practical option set, not every possible product. Inspirations identify design patterns. They do not imply feature equivalence or endorsement.

A Lean custom deployment

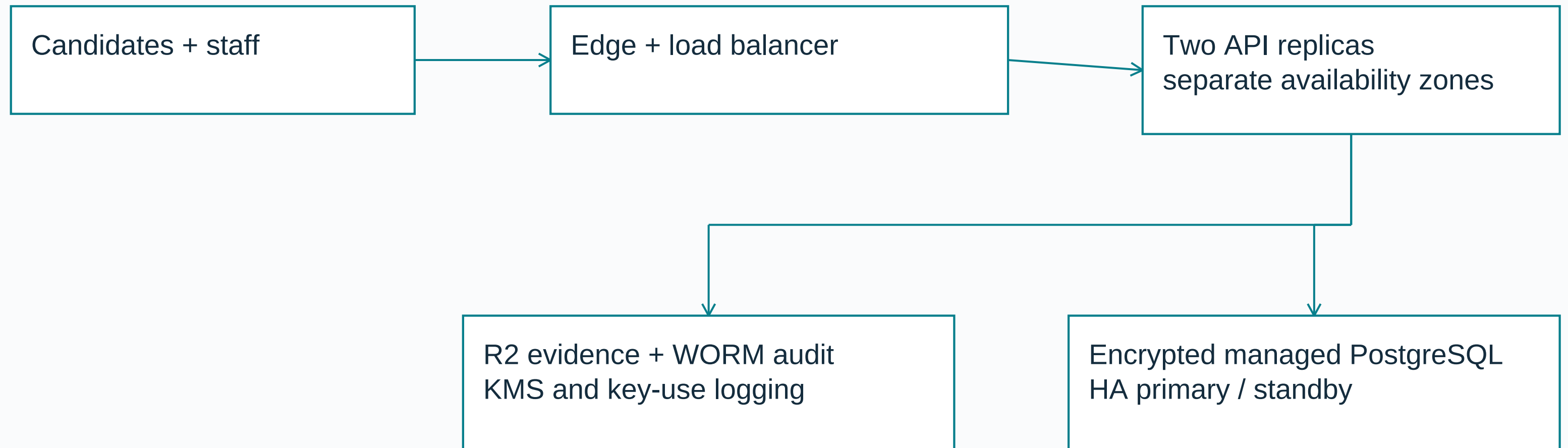


Fits a controlled pilot or an exam where restart interruption is acceptable.

A process or host failure stops admission and saves until recovery. Backups do not provide automatic failover.

Sizing is a starting configuration for a load test. Full 500-candidate capacity remains unproven. Monthly platform budget: \$124.85.

B Managed custom deployment

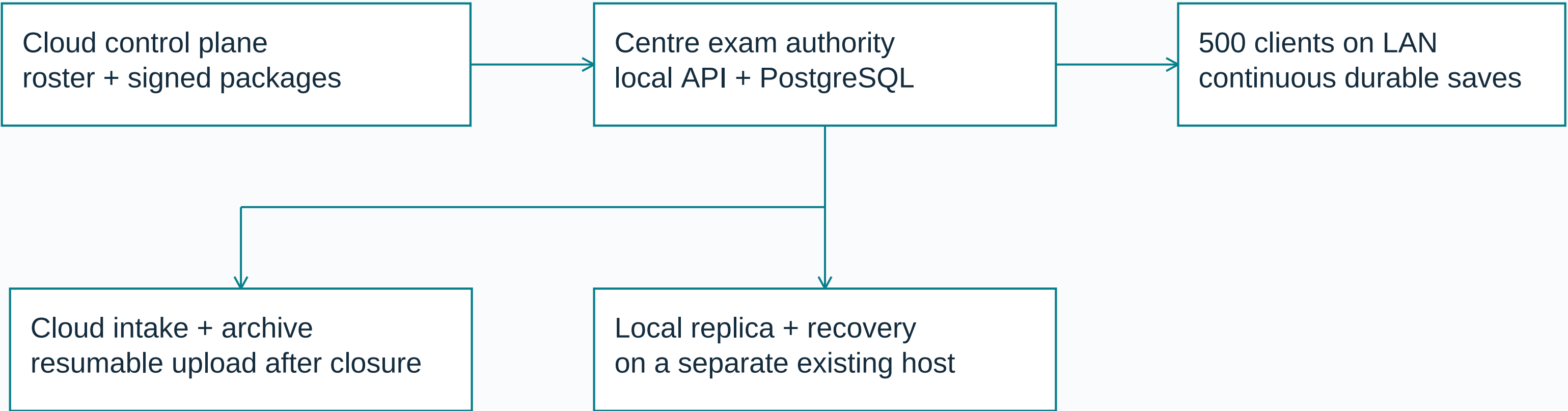


Use PostgreSQL-backed sessions and transactional admission. Redis is optional, not a mandatory extra service.

Both API replicas must survive independently. Move all process-local state into durable shared services.

Load balancer + 2 x 8 GB app VMs + encrypted 4 GB HA database. This improves service continuity, but does not protect against a regional or centre internet outage. Budget: \$305.45/month.

C Centre LAN during the exam, cloud sync afterwards

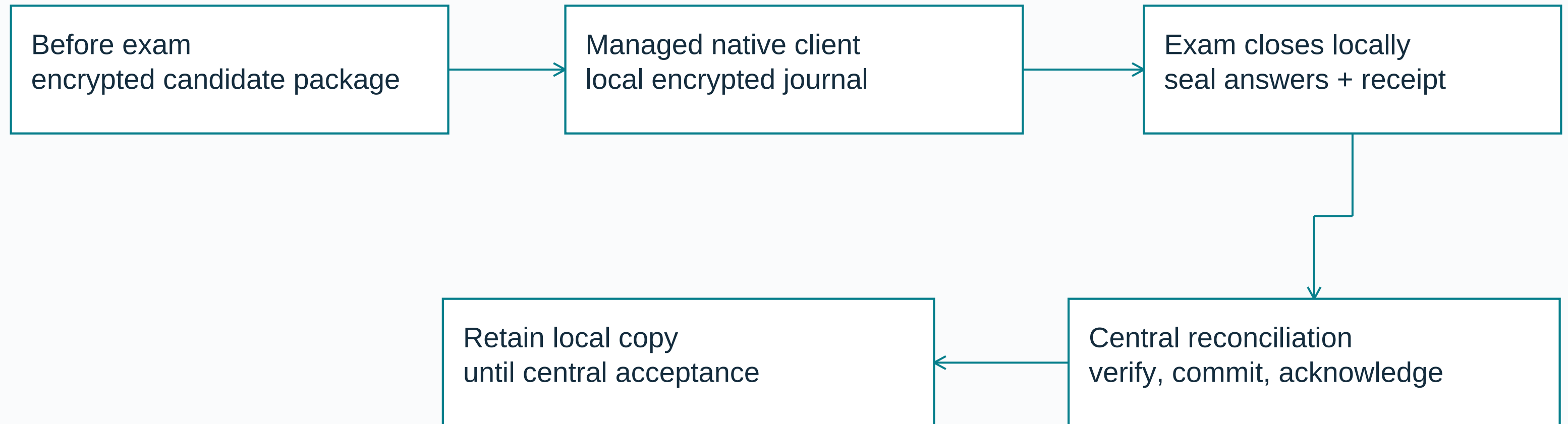


A signed centre permit authorises a fixed roster, paper version and exam window before internet disconnection.

A fencing rule selects one local authority. Never let a cloud and local writer independently own the same attempt.

Proposed design, not implemented. Existing local compute is an explicit zero-incremental-hosting assumption. Local software administration is included. Budget: \$93.65/month.

D Offline workstation, upload after the exam



Inspired by Exemplify's pending-upload workflow. This is a proposed custom design, not an ExamSoft implementation.

No live central revocation or invigilation while offline. Loss of a workstation can lose an entire unsynced attempt.

Use a native secured store with append-and-flush writes. A service worker or browser localStorage alone does not provide the required high-stakes trust. Budget: \$93.65/month.

End-of-exam sync is a verified import protocol

1 Issue a signed exam permit

Bind candidate, attempt, device, paper hash, allowed duration and a unique nonce. Keep the answer key off the workstation.

2 Record a durable journal

Append sequence-numbered answer events. Hash-chain entries and flush before displaying local-save success. Protect key access and detect rollback.

3 Close locally

Seal a final answer hash, journal root, final sequence and permit ID. Freeze editing. Display "completed locally, upload pending".

4 Upload in resumable chunks

Authenticate separately from the expired exam session. Use chunk hashes, bounded retries and an idempotent attempt/package key.

5 Reconcile and acknowledge

Verify signatures, membership, sequence and deadline evidence. Transactionally accept one result. Quarantine conflicts. Delete local data only after central acceptance.

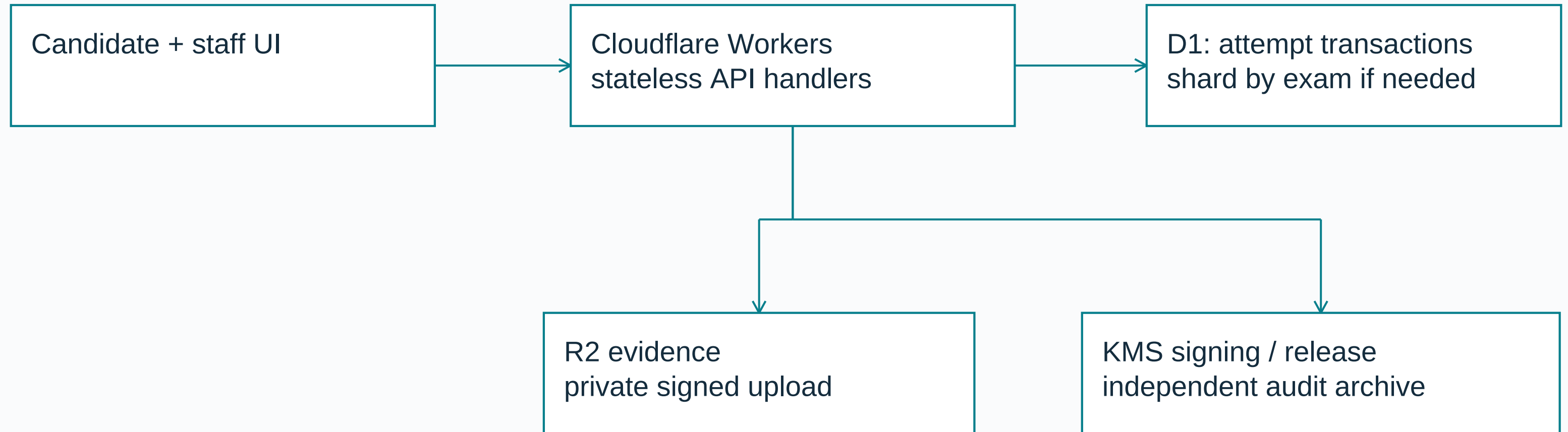
Offline signatures prove key possession, not honest timing on a compromised endpoint. Prefer a trusted centre clock and authority, or require human review when timing evidence is uncertain.

Batch sync changes the traffic shape, not the evidence volume

Payload / constraint	Calculation for one 500-seat session	Design response
Final answers + journal	Assume 1 MB/candidate = 500 MB	Upload first. At 20 Mbps effective uplink, theoretical minimum is 200 seconds.
Snapshot evidence	180 frames x 100 kB x 500 = 9 GB	At 20 Mbps, theoretical minimum is 3,600 seconds (60 minutes).
Both payloads	9.5 GB total = 76 gigabits	At 20 Mbps: 63m 20s. At 100 Mbps: 12m 40s, before overhead and retries.
500 simultaneous uploads	Burst can overload the network, API or database	Jitter start times, cap concurrency, resume by chunk and separate evidence from answer acceptance.
Extended outage	Central results remain pending	Retain encrypted local copies, reconcile 500 expected receipts, support controlled staff recovery.

Assumes decimal MB/GB and effective application bandwidth. Uploading once does not remove storage charges or provide a backup during the exam.

E Serverless custom platform

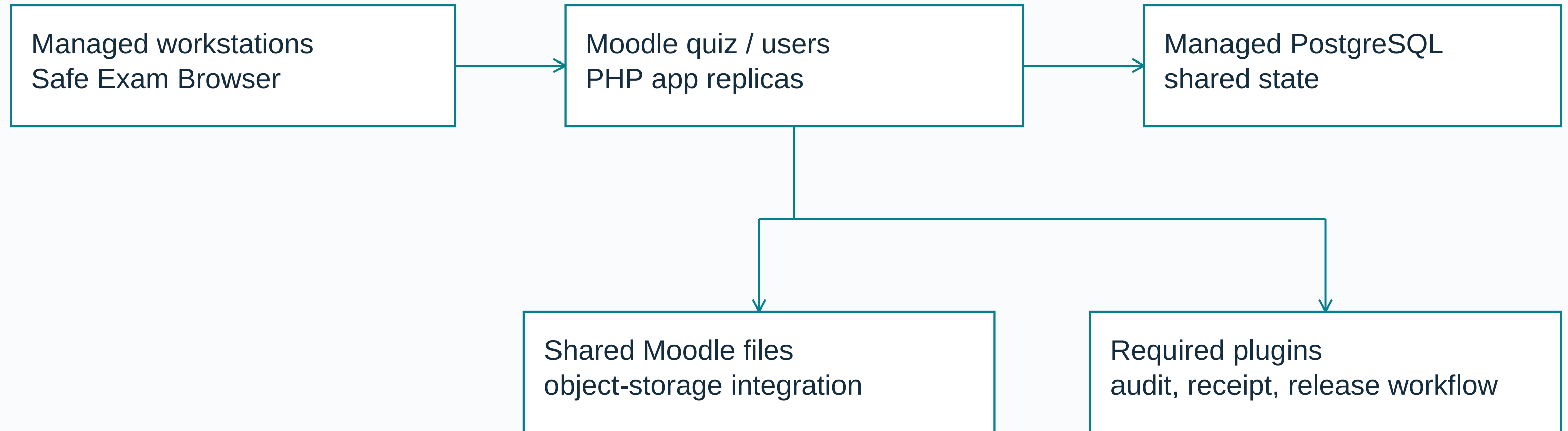


Rewrite Fastify and map-based services for the Worker runtime. Keep the same ownership, version and receipt rules.

D1 processes each database's queries serially and limits each database to 10 GB. Test write bursts, indexes and partitions.

8 GB total D1 storage is a budget input. External key-service latency and data location may dominate. A low monthly bill does not offset an unpriced rewrite. Budget: \$64.10/month.

F Moodle with Safe Exam Browser



Moodle provides quiz timing, navigation and randomisation options. SEB integrates with quiz access controls.

Signed paper manifests, custom publication approvals and this app's receipt format require an assessed integration gap.

Software licence cost assumed \$0 for open-source Moodle/SEB. Use supported shared storage and sessions, then benchmark this separate stack. No complete offline exam is assumed. Budget: \$305.45/month.

G Hosted examination service

Provider operates the exam platform

TestInvite is the priced example. Keep the client's identity, content and result-export requirements in the procurement checklist.

Different workflow and control ownership

Pro advertises API access, network restrictions and live monitoring. Confirm separation of duties, receipts, retention and export completeness.

Security features can change the bill sharply

Base test, webcam photos and lockdown consume different credits. A seat count alone is not enough to price usage.

500 simultaneous candidates require written acceptance

A question limit, annual credit count or "unlimited" plan does not establish tested concurrency, exam duration or an uptime commitment.

Offline is a separate procurement decision

Ask an offline-capable vendor for package, recovery and reconciliation terms. Do not assume TestInvite supports option D.

No supplier has quoted or committed capacity for this project. Published plans support conditional budget calculations, not a binding procurement total.

Tradeoffs that clients can choose explicitly

Option	Internet outage	Security tradeoff	Management / change burden
A	Exam interrupted	Single-host compromise and outage affect all state	Small stack, team owns recovery
B	Exam interrupted	Better backend continuity, still depends on endpoints	Managed DB reduces routine work
C	Exam continues on LAN	Local authority and physical access become critical	Local patching, replication and sync
D	Exam continues offline	Largest period without central control or backup	Native updates and disputed-upload handling
E	Exam interrupted	Provider isolation, plus region and partition constraints	Low VM work, specialised code
F	Exam interrupted	Established quiz flow, bespoke controls require review	Plugin compatibility and upgrades
G	Vendor-specific	Provider controls implementation and evidence export	Least platform work, contract dependence

These are relative design judgments. High availability, encryption, anti-cheating and privacy solve different risks and should not collapse into one security score.

Cost basis: a reproducible budget, not a fixed invoice

Published prices

Checked 8 September 2026. USD, monthly list pricing unless annual terms are shown. No promotional credits or reserved-instance discounts.

Shared usage

8 x 500 sittings/month, 90-day snapshot retention, existing domain, existing local compute for C, no equipment charges.

Service inclusions

Use recurring Cloudflare plan allowances on a dedicated account. KMS symmetric calls fit its allowance. No extra ingress or app transfer overage under the stated volumes.

Management assumptions

USD 25/hour planning rate. Routine maintenance varies by design. All options include 4 support hours/session: 1 preflight + 3 exam hours.

Excluded from recurring cost

Software development, hardware, OS/MDM licences, facilities, invigilators, biometric matching, SMS, taxes, FX and premium support contracts.

Logging reserve is an estimate, not a supplier quote. Annual figures are 12 x steady-state monthly budget, not a prediction of the first invoice. Unknown scope cannot be priced accurately as a fixed total.

Shared monthly services for custom options A-E and Moodle F

Service	Quantity and rate	Monthly USD
Edge plan [S5]	1 Cloudflare Pro zone, billed monthly	\$25.00
Snapshot objects [S6]	216 GB-month - 10 included = 206 x \$0.015	\$3.09
Object operations [S6]	Budget 0.80M class A / 0.10M class B, within included limits	\$0.00
Independent WORM audit [S7-S8]	10 GB x \$0.023 + 10k PUT x \$0.005/1k + 10k GET x \$0.0004/1k	\$0.28
Key service [S9]	2 keys x \$1 + 5k ECC signatures x \$0.15/10k; 10k symmetric calls included	\$2.08
Shared service subtotal	Sum rounded service line items	\$30.45
Operational logging reserve	Budget allowance for retained logs, alerts and small usage overruns	\$20.00

R2 evidence is private, not claimed immutable. Audit uses independent S3 Object Lock. Assume <=1 GB audit retrieval within AWS's unused 100 GB transfer allowance. Raw KMS cost is \$2.075 and audit is \$0.284.

Hosting bill of materials for each custom alternative

Option	Compute, database, backups and staging	Shared + logs	Total/month
A	VM \$48 + daily backup \$14.40 + staging \$12	\$50.45	\$124.85
B	2 app VMs \$88 + LB \$18 + HA DB \$120 + 100 GB snapshots \$5 + staging \$24	\$50.45	\$305.45
C	Cloud portal \$24 + daily backup \$7.20 + staging \$12. Existing LAN runtime \$0 incremental	\$50.45	\$93.65
D	Cloud import portal \$24 + daily backup \$7.20 + staging \$12. Local client hosting \$0 incremental	\$50.45	\$93.65
E	Workers \$5 + CPU \$1.40 + D1 storage \$2.25 + isolated staging allowance \$5	\$50.45	\$64.10
F	Same infrastructure allowance as B. Open-source licences \$0. Plugin development excluded	\$50.45	\$305.45

Staging holds synthetic data only. C/D cloud VMs process controlled post-exam imports, not 500 live exams. VM logs and database volume growth must fit the budgeted disk or be repriced.

Monthly hosting and app management comparison

Option	Platform	Routine ops	Exam support	Monthly total	Annualised
A Lean custom	\$124.85	16h / \$400.00	32h / \$800	\$1,324.85	\$15,898.20
B Managed custom	\$305.45	12h / \$300.00	32h / \$800	\$1,405.45	\$16,865.40
C Centre LAN	\$93.65	24h / \$600.00	32h / \$800	\$1,493.65	\$17,923.80
D Offline workstation	\$93.65	32h / \$800.00	32h / \$800	\$1,693.65	\$20,323.80
E Serverless custom	\$64.10	16h / \$400.00	32h / \$800	\$1,264.10	\$15,169.20
F Moodle + SEB	\$305.45	12h / \$300.00	32h / \$800	\$1,405.45	\$16,865.40
G Hosted service	\$5,116.67	8h / \$200.00	32h / \$800	\$6,116.67	\$73,400.00

G uses the published credit-purchase configuration with webcam photos + lockdown, not a confirmed 500-seat contract. Development and unquoted security add-ons are excluded. No guarantee of price or capacity.

Hosted service cost: feature selection changes credits

Configuration	Annual calculation	Annual / monthly equivalent
G base test only	Pro \$3,900 + 50,000-credit pack \$12,500. Need 48,000 tests less 500 included. 2,500 credits remain.	\$16,400 / \$1,366.67
G webcam photos + lockdown	6 credits/test x 48,000 = 288,000. Buy 287,500 x \$0.20 + Pro \$3,900.	\$61,400 / \$5,116.67
Potential Premium+ route	Page advertises \$35,000/year with unlimited credits. Confirm the same features, fair use, 500 concurrency and contract terms.	\$35,000 / \$2,916.67, conditional
Management for any G route	8 routine hours + 32 exam-support hours, at assumed \$25/hour. Add to the plan.	\$12,000 / \$1,000.00

Published credit arithmetic is not an offer. Confirm arbitrary pack size, validity, renewals and taxes. The 50k pack is cheaper than buying 47,500 credits at the smaller tier. Photo cadence may differ from custom options.

Sensitivity: the variables that move the budget

Change	Calculated impact	Meaning for the client
2 / 8 / 20 sessions per month	1,000 / 4,000 / 10,000 sittings. Scheduled app support: \$200 / \$800 / \$2,000 monthly	Support scales with exam days, even if servers stay the same
Routine labour rate: \$15 / \$25 / \$40	B has 44 total hours: \$660 / \$1,100 / \$1,760 per month, plus \$305.45 platform	A management contract matters more than a small VM discount
Snapshots off / every 60s / every 30s	Steady-state R2: \$0 / \$3.09 / \$6.33 storage. At 30s, roughly 1.6M class A requests adds \$4.50	Evidence ingestion and review grow with frequency
E: 20 ms / 100 ms mean Worker CPU	5M requests: CPU overage \$1.40 / \$9.40, before other services	Benchmark actual CPU, including authentication and crypto
D compared with B	Cloud saves \$211.80/month. Assumed extra routine work costs \$500/month	D is \$288.20/month higher under this management model

Frequency examples change support only, not a full reprice. More exams also increase storage and SaaS credits. R2 bills excess operations in whole-million units. Offline reduces network dependence, not necessarily cost.

Security additions are separate, explicit budget choices

Addition	How to price it	Decision / limitation
Face liveness at admission	Illustration: 4,000 sittings x 1.1 checks x \$0.015 = \$66/month in US East [S21]	Liveness is not identity matching. Regional availability, retries and integration need verification
Continuous biometric analysis	Quote service rate x frames analysed. Base model has 720,000 frames/month	Human observation and snapshot storage do not equal an automated face-matching service
Enterprise device trust	Quote certificate service, attestation and endpoint software separately	Existing device records do not establish real mTLS. Licences are excluded from this hosting scope
Dedicated HSM custody	Provider- and region-specific quote, plus redundant instances and administration	Use only where a contractual custody requirement justifies it. Managed KMS is the base budget
Independent assurance / premium support	Separate penetration-test and support quotes. Add recurring retests to management contract	No penetration-test, certification or 24/7 support fee is hidden inside a small cloud total

These are not included in A-G totals. A final Maximum Assurance quote must include every required addition. Staffing for evidence review and appeals is outside app management.

Operations and recovery objectives

Failure	Required handling	Proposed acceptance target
API process / replica loss	B reroutes to healthy replica. A restores service. Verify saved-answer consistency	B: resume within 60s, zero acknowledged-answer loss for a tested replica failure
Database failover	Transaction retry, pooled connections, duplicate-safe writes	B/C: target <=5 min recovery, verify acknowledged writes under the selected replication mode
Centre internet loss	C uses local authority. D seals locally and waits. A/B/E/F need internet restoration	C/D: finish a full 3h exam offline in rehearsal
Local disk / host failure	C switches to fenced replica. D uses approved recovery copy where available	D cannot promise zero data loss without an independent durable copy
Regional outage or restore	Independent backup, keys and restore procedure, exam-controller decision	Recovery target requires a separate DR design. No automatic cross-region recovery is priced

Targets are acceptance criteria, not measured SLAs. A failover plan must cover key access, sessions, evidence and DNS as well as database health.

Production plan and acceptance gates

1 Client decisions

Choose online, centre-LAN or workstation-offline mode. Set exam frequency, allowed outage, retention and identity requirements.

2 Durable foundations

Implement storage transactions, shared sessions, durable key custody and an external audit checkpoint. Remove demonstration data and bypasses.

3 Delivery-specific controls

Implement real endpoint trust or the offline permit/journal protocol. Complete retention jobs, monitoring and support runbooks.

4 Full rehearsal

Run 500 candidates for three hours plus staff. Test starts, answer bursts, final submission, browser restarts, network loss and restore.

5 Security and go-live acceptance

Independent security assessment. Resolve failed controls. Confirm invoice-region pricing, supplier terms and staff escalation ownership.

Go-live gates: no lost acknowledged answers, no answer-key leakage, duplicate-safe reconciliation, 500 expected receipts accounted for, and documented handling of every rejected upload.

Client decision worksheet

Decision	Recommended starting position	What changes if the client chooses otherwise
Internet dependence	C if the centre must continue through WAN failure, otherwise B	D adds full workstation autonomy and more endpoint risk
Identity	Supervised admit-card check, with optional biometric integration	More biometric assurance adds provider cost and privacy work
Evidence	Snapshots only where necessary, explicit 90-day planning retention	Video, legal holds or longer retention increase storage and review scope
Management	Named app operator plus scheduled exam support	24/7 response needs a different staffing contract
Data location	Client confirms approved regions before procurement	India-only storage may rule out or reprice the global reference designs
Workflow ownership	Keep custom workflow if paper custody and receipts are essential	Moodle/SaaS reduce custom work if the client accepts a fit-gap outcome

The lowest recurring hosting price is E. The more balanced shortlist is B/C. Select D for network independence, not as an assumed cost-saving shortcut.

Appendix: source-backed implementation status

Implemented in the POC	Partial or simulated	Required before live high-stakes use
Password hash, cookie signature, role permissions	Secure cookie / HSTS depend on production deployment	Staff MFA, hardened proxy path, shared session persistence
Canonical hashes, manifest signatures, AES-GCM	Ephemeral development keys and application release gate	Durable KMS custody, authorised release, key rotation and restore
Ownership, assignment checks, versions, idempotency	Map-backed answer/event storage and in-memory audit chain	Transactional persistence, external immutable audit and tested recovery
Capture challenges, evidence hashes, warning workflow	Mock face, fingerprint and device-health verdicts	Real provider / native-agent integration if the selected policy needs it
Certificate metadata and registration	mTLS, kiosk, TPM attestation and synthetic DDoS scenarios	Deploy and test actual network and endpoint enforcement
Retention fields and recorded access	No proven production lifecycle or backup deletion workflow	Retention scheduler, evidence retrieval controls, deletion verification

Implemented means a real code path, not a certified or independently verified security claim. Default configuration may leave optional controls disabled.

Appendix: cost formulas and non-equivalent features

Formula / condition	Application
Monthly total = platform + routine hours x rate + sessions x support hours x rate	All A-G totals use rate \$25/hour, 8 sessions and 4 support hours/session.
Annualised = monthly total x 12; cost/sitting = monthly total / 4,000	B: \$1,405.45 monthly, \$16,865.40 annualised, \$0.35 per sitting.
Evidence = 500 x 8 x 180 x 100,000 bytes	72 GB new data/month. At three months retention: 216 GB steady-state planning storage.
E CPU = max(5M x 20ms - 30M ms, 0) / 1M x \$0.02	CPU \$1.40. D1 8 GB less 5 GB included x \$0.75 = \$2.25. Budget 10M written and 100M read rows, within paid allowances.
SaaS credit total depends on enabled features	G photos and lockdown do not prove the same cadence, custody or offline capabilities as the custom alternatives.
Quote-required items remain unknown, not zero	Development, regional restrictions, native client licensing and enterprise security must enter the final commercial scope.

Prices are rounded at service-line level, then summed. Provider invoice rounding, tax and actual measured consumption can differ. No equipment prices appear in this model.

Appendix: repository evidence

R1 Repository overview and design

README.md; docs/architecture.md; docs/limitations.md

R2 Security controls and policy

docs/security.md; packages/shared/src/securityProfiles.ts

R3 Runtime and storage

apps/api/src/app.ts; config/env.ts; lib/session.ts; lib/store/adapters.ts; lib/store/db.ts

R4 Paper and answer integrity

apps/api/src/lib/crypto/paper.ts; lib/crypto/keyProvider.ts; services/answerService.ts; services/attemptService.ts

R5 Offline and candidate flow

apps/web/src/hooks/useAnswerSaver.ts; lib/candidateStore.ts; pages/candidate/*

R6 Evidence and human review

apps/api/src/services/monitoringService.ts; apps/web/src/components/domain/CameraMonitor.tsx

Paths are relative to the exam repository. For R3/R4 entries after the first path, use apps/api/src/. For R5 entries, use apps/web/src/.

Sources: hosting and storage

S1 DigitalOcean Droplet prices

<https://www.digitalocean.com/pricing/droplets>

S2 DigitalOcean backup prices

<https://www.digitalocean.com/pricing/backups>

S3 Amazon Lightsail prices

<https://aws.amazon.com/lightsail/pricing/>

S4 Lightsail snapshot billing

<https://docs.aws.amazon.com/lightsail/latest/userguide/amazon-lightsail-faq-snapshots.html>

S5 Cloudflare Pro monthly plan

<https://www.cloudflare.com/plans/>

S6 Cloudflare R2 pricing and rounding

<https://developers.cloudflare.com/r2/pricing/>

S7 Amazon S3 pricing

<https://aws.amazon.com/s3/pricing/>

S8 AWS example confirming S3 storage and PUT rates

<https://docs.aws.amazon.com/solutions/latest/live-streaming-on-aws-with-amazon-s3/cost-example-1.html>

S9 AWS KMS pricing and signing example

<https://aws.amazon.com/kms/pricing/>

Primary vendor sources checked 8 September 2026. Regional and contract prices require revalidation before purchase.

Sources: alternative platforms

S10 Cloudflare Workers pricing

<https://developers.cloudflare.com/workers/platform/pricing/>

S11 Cloudflare D1 pricing

<https://developers.cloudflare.com/d1/platform/pricing/>

S12 Cloudflare D1 capacity limits

<https://developers.cloudflare.com/d1/platform/limits/>

S13 Moodle Safe Exam Browser integration

https://docs.moodle.org/502/en/Safe_Exam_Browser

S14 Moodle quiz settings

https://docs.moodle.org/502/en/Quiz_settings

S15 Safe Exam Browser overview and licence

https://safeexambrowser.org/about_overview_en.html

S16 TestInvite plans and credit calculator

<https://www.testinvite.com/lang/en/pricing.html>

S17 ExamSoft offline submission behaviour

<https://support.examsoft.com/hc/en-us/articles/47010017980941-How-to-Submit-Your-Exam-With-Video>

S18 CouchDB replication model

<https://docs.couchdb.org/en/stable/replication/intro.html>

Sources: security and usage allowances

S19 S3 Object Lock

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

S20 AWS internet transfer allowance

<https://aws.amazon.com/about-aws/global-infrastructure/global-network/faqs/>

S21 Rekognition liveness pricing

<https://aws.amazon.com/rekognition/pricing/>

S22 Rekognition liveness limitations

<https://docs.aws.amazon.com/rekognition/latest/dg/face-liveness.html>